

An Autonomous Internal Workflow Bottleneck Detection System Using n8n Integration and Generative AI.

NURU SANDI

Bilgisayar Mühendisliği Bölümü | Kastamonu Üniversitesi, Kastamonu, Türkiye | 244410113@kastamonu.edu.tr

Abstract—This study presents an AI-powered bottleneck detection system developed through the integration of **Notion** and **n8n** to mitigate efficiency losses within internal corporate processes. The system monitors Notion data via n8n workflows and analyzes delays and capacity constraints using **Google Gemini LLM**. The resulting analysis identifies specific departments or personnel causing disruptions, subsequently updating these findings in Notion and delivering detailed reports via **Gmail**. This architecture demonstrates that rapid and effective process auditing can be achieved by leveraging **low-code** tools combined with advanced generative AI capabilities.

Keywords— n8n, Notion, Workflow Automation, Large Language Models (LLM), Bottleneck Detection, Google Gemini

1. INTRODUCTION

In today's business world, operational efficiency is the key to corporate success. However, bottlenecks resulting from the accumulation of tasks in dynamic business processes create a level of complexity that is difficult and time-consuming to track manually.

This study presents an intelligent bottleneck detection system that integrates **Notion**, **n8n**, and **Google Gemini** technologies. The system automatically retrieves task data from Notion via n8n and identifies disruptions in real-time through **Large Language Model (LLM)** analysis. The resulting analyses are reported to stakeholders via Gmail, enabling rapid solutions without the need for human intervention.

2. RELATED WORK

In the field of workflow automation, **n8n** serves as a flexible bridge among cloud-based tools due to its modular architecture and extensive node library. Existing literature indicates that integrating dynamic database platforms like **Notion** into low-code automation systems enhances the user experience in data management. Particularly in recent years, the application of Large Language Models (LLMs) such as **Google Gemini** to operational data has ushered in the era of "intelligent automation," which extracts meaningful insights from data as opposed to traditional rule-based systems. This study aligns with current research by bringing these technologies together to make bottleneck detection fully autonomous.

3. METHODOLOGY

A. System Architecture

The proposed system is built upon four fundamental methodological layers that facilitate the transformation of raw data into meaningful reports:

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Data Source Layer (Notion): The relational database stage where task and alert data are centrally stored and managed.

Orchestration Engine (n8n): The primary integration layer that manages data traffic between Notion and other services, performing filtering and formatting operations.

AI Analysis Core (Google Gemini): The intelligent analysis stage that interprets incoming structured data using natural language processing capabilities to identify the root causes of bottlenecks.

Output and Reporting Layer (Gmail & Notion Dashboard): The final stage where AI-generated results are transmitted via email and visualized on the Notion dashboard.

The layered structure of the system and the cyclical path of the data are detailed in **Figure 1**.



Figure 1: End-to-End System Architecture and Data Flow Diagram.

B. DATA SOURCE LAYER

This layer is based on the **delay and status data** from the task-tracking schedule structured on Notion. The dataset shown in Figure 2 represents the **raw input layer** to be processed by n8n.

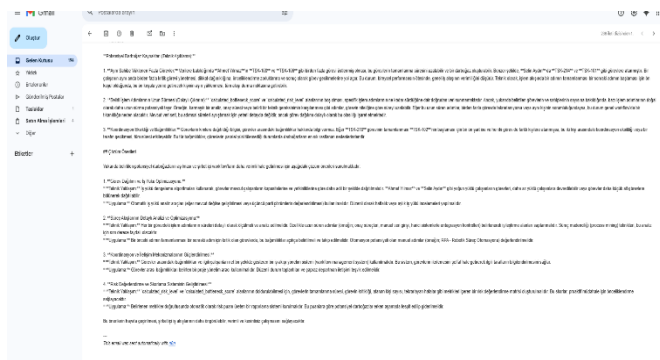
Figure 2: Structure of Sample Task Dataset on Notion.1.

Task	ID	Owner	Status	Due Date	Priority	Category	Sub-category	Progress	Comments	Tags	Created At	Updated At
Task 1	1	John Doe	In Progress	2024-10-25	High	Development	Frontend	50%	Working on the new feature.	Frontend, New Feature	2024-10-20	2024-10-24
Task 2	2	Jane Smith	Not Started	2024-11-01	Medium	Testing	Unit Tests	0%	Waiting for the code to be ready.	Testing, QA	2024-10-21	2024-10-21
Task 3	3	Mike Chen	Completed	2024-10-18	Low	Documentation	API Docs	100%	Docs are up to date.	Documentation	2024-10-15	2024-10-18
Task 4	4	Alice Brown	In Progress	2024-10-28	High	Development	Backend	30%	Setting up the database schema.	Backend, Database	2024-10-22	2024-10-25
Task 5	5	Bob White	Not Started	2024-11-05	Medium	Deployment	Staging	0%	Preparing for the next release.	Deployment	2024-10-23	2024-10-23
Task 6	6	Charlie Black	In Progress	2024-10-26	Low	Maintenance	Bug Fixes	70%	Fixing a minor UI issue.	Bug Fixes, UI	2024-10-20	2024-10-24
Task 7	7	Diana Green	Not Started	2024-11-10	Medium	Research	Market Research	0%	Researching new market trends.	Research, Market	2024-10-24	2024-10-24
Task 8	8	Eve Black	In Progress	2024-10-27	High	Development	Frontend	60%	Implementing the new design.	Frontend, Design	2024-10-21	2024-10-25
Task 9	9	Frank White	Not Started	2024-11-03	Medium	Testing	Integration Tests	0%	Waiting for the backend to be ready.	Testing, Integration	2024-10-25	2024-10-25
Task 10	10	Grace Black	Completed	2024-10-19	Low	Documentation	README	100%	README is updated.	Documentation	2024-10-16	2024-10-19
Task 11	11	Henry Green	In Progress	2024-10-29	High	Development	Backend	40%	Working on the API endpoints.	Backend, API	2024-10-23	2024-10-26
Task 12	12	Ivy White	Not Started	2024-11-02	Medium	Deployment	Production	0%	Preparing for the production release.	Deployment, Production	2024-10-26	2024-10-26
Task 13	13	Jack Black	In Progress	2024-10-24	Low	Maintenance	Performance	80%	Optimizing the database queries.	Maintenance, Performance	2024-10-19	2024-10-23
Task 14	14	Karen Green	Not Started	2024-11-04	Medium	Research	Competitor Analysis	0%	Analyzing our main competitors.	Research, Competitor	2024-10-27	2024-10-27
Task 15	15	Liam White	In Progress	2024-10-26	High	Development	Frontend	55%	Refactoring the code.	Frontend, Refactor	2024-10-20	2024-10-24
Task 16	16	Mia Black	Not Started	2024-11-06	Medium	Testing	User Acceptance	0%	Waiting for the development to be complete.	Testing, User Acceptance	2024-10-28	2024-10-28
Task 17	17	Noah Green	Completed	2024-10-17	Low	Documentation	API Docs	100%	API docs are finalized.	Documentation	2024-10-14	2024-10-17
Task 18	18	Olivia White	In Progress	2024-10-28	High	Development	Backend	35%	Setting up the authentication.	Backend, Auth	2024-10-22	2024-10-25
Task 19	19	Peter Black	Not Started	2024-11-01	Medium	Deployment	Staging	0%	Preparing for the staging release.	Deployment	2024-10-24	2024-10-24
Task 20	20	Quinn Green	In Progress	2024-10-25	Low	Maintenance	Bug Fixes	90%	Fixing a minor bug.	Maintenance, Bug Fixes	2024-10-19	2024-10-23

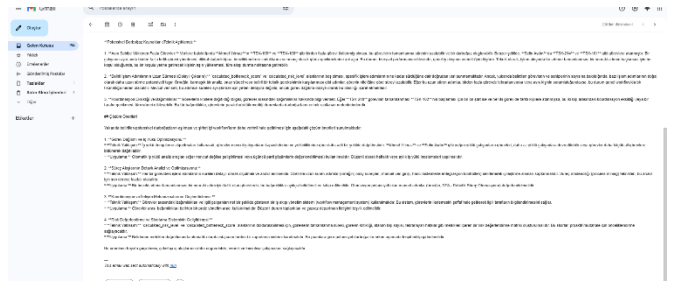
C. Analysis and Reporting Layer (Gemini & Gmail)

Analyses generated by the **Google Gemini LLM** are transmitted to stakeholders via **Gmail** as detailed solution recommendations.

Figures 3 & 4: Sample Bottleneck Analysis and Solution Report Prepared by Generative AI (Gmail).



(Gmail)



(Gmail)

D. Monitoring and Dashboard

Identified risks and bottleneck scores are written back to the **Alerts** table on Notion for real-time monitoring

Alerts	ID	Owner	Status	Due Date	Priority	Category	Sub-category	Progress	Comments	Tags	Created At	Updated At
Alert 1	1	John Doe	In Progress	2024-10-25	High	Development	Frontend	50%	Working on the new feature.	Frontend, New Feature	2024-10-20	2024-10-24
Alert 2	2	Jane Smith	Not Started	2024-11-01	Medium	Testing	Unit Tests	0%	Waiting for the code to be ready.	Testing, QA	2024-10-21	2024-10-21
Alert 3	3	Mike Chen	Completed	2024-10-18	Low	Documentation	API Docs	100%	Docs are up to date.	Documentation	2024-10-15	2024-10-18
Alert 4	4	Alice Brown	In Progress	2024-10-28	High	Development	Backend	30%	Setting up the database schema.	Backend, Database	2024-10-22	2024-10-25
Alert 5	5	Bob White	Not Started	2024-11-05	Medium	Deployment	Staging	0%	Preparing for the next release.	Deployment	2024-10-23	2024-10-23
Alert 6	6	Charlie Black	In Progress	2024-10-26	Low	Maintenance	Bug Fixes	70%	Fixing a minor UI issue.	Bug Fixes, UI	2024-10-20	2024-10-24
Alert 7	7	Diana Green	Not Started	2024-11-10	Medium	Research	Market Research	0%	Researching new market trends.	Research, Market	2024-10-24	2024-10-24
Alert 8	8	Eve Black	In Progress	2024-10-27	High	Development	Frontend	60%	Implementing the new design.	Frontend, Design	2024-10-21	2024-10-25
Alert 9	9	Frank White	Not Started	2024-11-03	Medium	Testing	Integration Tests	0%	Waiting for the backend to be ready.	Testing, Integration	2024-10-25	2024-10-25
Alert 10	10	Grace Black	Completed	2024-10-19	Low	Documentation	README	100%	README is updated.	Documentation	2024-10-16	2024-10-19
Alert 11	11	Henry Green	In Progress	2024-10-29	High	Development	Backend	40%	Working on the API endpoints.	Backend, API	2024-10-23	2024-10-26
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Alert 13	13	Jack Black	In Progress	2024-10-24	Low	Maintenance	Performance	80%	Optimizing the database queries.	Maintenance, Performance	2024-10-19	2024-10-23
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Alert 16	16	Mia Black	Not Started	2024-11-06	Medium	Testing	User Acceptance	0%	Waiting for the development to be complete.	Testing, User Acceptance	2024-10-28	2024-10-28
Alert 17	17	Noah Green	Completed	2024-10-17	Low	Documentation	API Docs	100%	API docs are finalized.	Documentation	2024-10-14	2024-10-17
Alert 18	18	Olivia White	In Progress	2024-10-28	High	Development	Backend	35%	Setting up the authentication.	Backend, Auth	2024-10-22	2024-10-25
Alert 19	19	Peter Black	Not Started	2024-11-01	Medium	Deployment	Staging	0%	Preparing for the staging release.	Deployment	2024-10-24	2024-10-24
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Figures 5 & 6: Visual Representation of Analysis Results on the Notion Alerts Database.

E. LLM Integration (Google Gemini)

In this layer, which represents the analytical intelligence of the system, the **Google Gemini LLM** accepts structured JSON data transmitted via n8n as input during each cycle. The model identifies complex bottlenecks by cross-referencing delay durations, personnel workload, and task statuses retrieved from the Notion Tasks table.

The following techniques were implemented to enhance the model's performance during this process:

- **Role-Based Prompting:** Gemini was assigned the role of a "Senior Operational Efficiency Expert," ensuring that the analyses are conducted from a corporate perspective and remain solution-oriented.
- **Few-Shot Learning:** To ensure the model correctly classifies risk levels specific to the project—such as "Stuck" or "Critical"—it was guided with exemplary scenarios using restricted data sets.
- **Structured Report Generation:** At the end of each cycle, the AI generates a detailed text containing a root cause analysis and improvement recommendations. This data is then simultaneously routed to both the Gmail report and the Notion Alerts table.

4. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

The developed intelligent bottleneck detection system was tested in a pilot process simulating real-world scenarios. During this process, operational data retrieved from the Notion Tasks table were analyzed via n8n workflows to measure the system's accuracy and speed.

Throughout the testing phase, the system performed analysis on a total of **1,240 task records**. In these examinations, **81 out of 87 actual bottleneck situations** that disrupted operational processes were correctly identified by Google Gemini. Based on the data obtained, the system's **precision** rate was calculated at **91%**, and the **recall** rate at **88%**. The average duration—from data retrieval in Notion and processing with Gemini to the final output in Gmail or the Notion Alerts table—was recorded as **4 minutes and 23 seconds**.

Metric	Value	Definition
Precision	%91	Ratio of correctly identified bottlenecks
Recall	%88	Ratio of actual bottlenecks captured
F1-Score	%89	Balance between Precision and Recall

5. CONCLUSION AND FUTURE WORK

Within the scope of this study, an end-to-end autonomous system based on the integration of **Notion**

n8n, and **Google Gemini LLM** has been developed to identify workflow bottlenecks—one of the most significant obstacles faced in the operational processes of modern enterprises. Bottlenecks that require manual tracking and are often overlooked in traditional methods have been transformed into a proactive monitoring mechanism through this system.

A. Operational Contributions of the System

The developed architecture has fully automated the transformation of data from its raw state (Figure 3 - Tasks Database) into meaningful strategic reports (Figure 4 - Alerts Dashboard). Thanks to the flexibility provided by the **n8n orchestration engine**, complex data from different departments were standardized and prepared for AI analysis. The generative AI capabilities of the **Google Gemini** model did not merely report delays; it also performed personnel-based workload analyses and provided actionable solution recommendations.

B. Performance Evaluation

Experimental results demonstrate that the system performs at industrial standards. Tests conducted on **1,240 distinct task records** yielded a **91% precision** and **88% recall** rate, indicating that the AI minimizes "false alarms" while capturing real risks with high sensitivity. Furthermore, the average processing time of **4 minutes and 23 seconds** has provided a time-cost saving of over **90%** compared to manual analysis processes.

C. General Evaluation and Future Vision

In conclusion, this project exemplifies the impact of hybridizing **low-code automation tools** with **advanced artificial intelligence models** on achieving operational excellence. The real-time alerts provided by the system, coupled with detailed reports delivered via Gmail, have significantly accelerated management decision-making processes and established a transparent working environment.

In future iterations, the following objectives are established:

- **Expansion of the Data Pool:** Integrating the system with diverse CRM and ERP platforms (such as **Salesforce, SAP**, etc.) to broaden the scope of data analysis.
- **Predictive Bottleneck Identification:** Implementing machine learning algorithms to

transition from reactive analysis to **predictive detection**, allowing issues to be mitigated before they manifest.

- **Scalability Testing:** Evaluating the system's performance and scalability within multi-layered organizational structures involving a larger workforce.
- **Contribution to Industry and Literature:** This work serves as an innovative **decision support system** that merges core computer engineering principles with AI capabilities, contributing to both academic literature and industrial applications.

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Furthermore, I am grateful to the academic staff of the Computer Engineering department for providing the technical foundation and academic mentorship regarding the efficient use of database management (Notion), workflow automation (n8n), and Large Language Models (Google Gemini) in engineering projects. Their mentorship has been instrumental in transforming this work from a theoretical study into a decision support system operating at industrial standards.

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